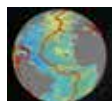




## Photos for password

With Passage, you can replace your passwords with an image held in front of your camera while logging in <http://dgit.in/PassagePass>



## Ocean Maps

With satellite imagery, researchers have developed an ocean map showing undersea mountains <http://dgit.in/OceanDepths>

**Touched By Tech**

Silva Cells are modular pavement system frameworks. They offer a structural soil alternative.

Each silva cell is made up of a frame and a deck. It's 48x24x16 inches in size and can hold about 10 cubic feet of soil. The structure is such that water drainage is easy. The cells can be layered on top of each other, thus forming a compact structure which allows trees the rooting zone and soil quality they need to grow.

Silva cells will make cities cleaner, greener and more sustainable. They'll help battle rising temperatures, improve air and water quality, reduce pollution, flooding and erosion.

## FishNET technology

Illegal, Unreported and Unregulated (IUU) fishing may not seem like they have a direct impact on the environment. But these crimes lead to ecosystem corruption, illegal dumping, pollution and trespassing on protected areas. The FishNET Project allows the concerned authorities to easily track offenders by concentrating on targeted regions. FishNET uses simple technology and communication methods. It depends on timely and accurate reporting. It is a web-based information system that captures and analyses crowd-sourced data on exploited areas and fishing vessels. With increased community participation, the FishNET database has been drastically improving leading to effective enforcement of ocean policy and security. It's a powerful tool to fight IUU fishing activities.

## Supercritical carbon dyeing

The textile industry uses lavish amounts of water to dye fabrics. Almost 25 litres is used to dye a single T-shirt. Besides this water wastage, most textile plants are located in Eastern Asia, where the resulting chemically polluted water is

sometimes dumped into local rivers. A recently discovered technology has made it possible to substitute water with carbon dioxide. 'Applied Separations' is an organisation that manufactures this supercritical carbon dioxide which takes up the dye. A supercritical fluid is basically a highly compressed gas. It exhibits properties of both, gas and liquid – being able to move through solids as well as dissolve substances. Color dissolved in carbon dioxide penetrates fabrics easily. There's no need of salt, water or harmful chemicals. Since the fabric emerges from this process dry, the dyeing time is reduced to about half the normal amount. And of course, less carbon dioxide in the air is a major plus too.

## Salt water air conditioners

This is a new technology based on non-toxic, salt water solution. It dehumidifies and cleans the air indoors while saving energy. Conventional air conditioners chill the air to its dew point and then reheat it to a comfortable temperature. By passing air through a 'liquid desiccant' i.e. a non-toxic, fluid, salt water solution, this process dehumidifies air without the need for reheating. It also reduces energy consumption by up to 40 percent. This process naturally disinfects the surrounding air freeing it from micro-organisms and bacteria. These ACs are currently being used in Hawaii, Toronto and Sweden.

## Solar roadways

Imagine a smart highway with solar-powered road panels instead of the regular asphalt and concrete. A road that could generate renewable energy and, in

the long-term, pay for itself. This idea can be extended to parking lots, driveways, footpaths, houses etc. The solar roadways consist of greenish hexagonal tiles. Each of these tiles has a solar cell with a sheet of hardened glass above it. The roadways could even be used to power electric cars with the excess energy generated.

## Zooniverse

'Zooniverse' is a citizen science web portal headquartered at Oxford. It has 1.1 million volunteers called 'Zooites'. The projects are both, environmental and scientific. The 'zooites' perform computing to complete research tasks. This crowd-sourced scientific research is used to keep tabs



We'll soon have solar-powered highways

on data pertaining to the environment, from melting ice in the Arctic regions to detecting oncoming cyclones.

For example, 'Plankton Portal', one such Zooniverse project classifies plankton distribution at different ocean depths from images gathered by the In Situ Ichthyoplankton Imaging System (ISIIIS). Using plankton as a valid indicator, ocean carbon dioxide levels can be diagnosed and we can be alerted to unusual rises in carbon dioxide.



H91M-PLUS H91M-S1 PLUS H81M-S1 PLUS

# Best Wishes and Greetings for Diwali

## FREE Gifts Exclusively from ASRock

Limited Period : October–November, 2014

**ASRock**  
No. 1 in Motherboards